



BlockDNS Whitepaper

A lifetime, on-chain naming system. .bdns domains, IPFS hosting, multi-wallet identity and a token economy that funds the chain — native on the BlockDNS L2.

Protocol demo running live on the local BlockDNS L2 (chain 8461)

≡ Abstract

Internet navigation today rents you a name. Renewal fees, registrars, censorship and opaque pricing — ownership is never real. BlockDNS replaces DNS with an on-chain registry: a **.bdns** domain is an ERC-721 NFT minted once, held forever, and resolvable to IPFS content plus wallet bindings for ETH, BTC and SOL.

Registration is a single, on-chain action paid in BDNS — the protocol token. Common names (5+ characters) mint for free; short premium names are priced dynamically, burning 50% of the payment and routing 50% to the treasury. No subscriptions, no renewals, no rent.

🔑 The problem: you don't own your name

Rent, not ownership. Every traditional domain expires. Miss a renewal and your identity, email and brand can be hijacked.

Centralized control. Registries, governments or quirks of a single corporation can seize or censor a name at any layer.

One name, many islands. Your website, crypto addresses and social handles live in disconnected silos — proving ownership of a brand is manual and fragile.

🏗️ Solution: BlockDNS L2

BlockDNS runs as a fast, low-cost L2 chain (chain ID 8461). Every domain, record and payment lives on-chain — auditable forever and owned by a single wallet.

The stack keeps the familiar mental model of DNS while replacing the authority: a **registry** contract mints and owns domains, a **pricer** sets dynamic fees, the **BDNS** contract is the store of value, and a **marketplace** lets holders trade names in a trustless escrow.

🌐 How .bdns domains work

A domain is an **ERC-721 NFT**. Minting burns the chosen name into the registry forever — no two wallets can own the same name.

The mint of a 5+ character name is **free**. Premium short names pay a one-time, on-chain dynamic price in BDNS.

Owners attach an **IPFS CID**: keccak name resolution serves content over any IPFS gateway — truly decentralized hosting, pinning-free to read.

Owners bind **wallet records** (ETH, BTC, SOL and any chain) to a single domain, turning it into a portable, battle-tested digital identity.

Ownership is liquid: a domain can be **sold on the marketplace** or transferred through standard ERC-721 primitives.

🔗 The BDNS token

Supply: hard cap of 1,000,000,000 (1B) BDNS, an ERC-20 on the BlockDNS L2.

Utility: payment for premium registrations and marketplace purchases; the accounting unit of the whole protocol.

Distribution: genesis issuance, protocol treasury and a community airdrop — all on-chain and auditable from day one.

🔗 Token distribution

1,000,000,000 BDNS is the absolute maximum supply. The allocation below is on-chain from genesis — no hidden mints, every wallet divisible and verifiable at any block.

ALLOCATION	SHARE	BDNS (OF 1B)	UNLOCK SCHEDULE
Ecosystem & Treasury	60%	600,000,000	10% at TGE, then linear over 24 months
Community Staking	10%	100,000,000	Claimable at 12 months · staking rewards (top 10)
Team & Advisors	15%	150,000,000	6-month cliff, 24-month vesting
Launch & Liquidity	5%	50,000,000	Locked at TGE, released with LPs
Marketing & Partnerships	5%	50,000,000	6-month linear vesting
Chain Reserve	5%	50,000,000	Reserved for protocol operations

📊 40% ecosystem

The largest share. Funds liquidity, integrations and chain operations for the long term.

👥 25% community

Airdrop to active users — the widest possible fair-launch distribution of the protocol.

🔗 15% team

Cliff-plus-vesting keeps builders aligned with the protocol for years, not days.

≡ How the token flows — mechanics

BDNS moves in full-circle, transparent loops. Every flow is a smart contract transfer — no off-chain ledger, no discretion.

1 · Free registration (5+ chars)

Mint costs **0 BDNS**. Creates a namespace asset at no cost — the grassroots flywheel of the protocol.

2 · Premium registration (short names)

A one-time on-chain price paid in BDNS: **50% is burned** (deflation) and **50% flows to the treasury**.

3 · Marketplace resale

When a domain sells, the buyer pays the listed price in BDNS; a **2.5% fee** is routed to the treasury and the seller receives the rest instantly.

4 · Airdrop claims

Eligible wallets claim their allocation from the Merkle contract; **unclaimed tokens roll back to the ecosystem treasury**.

5 · Treasury re-deployment

Accumulated BDNS returns to the ecosystem as liquidity, grants and chain incentives — completing the loop.

`mint(free) → supply static`

`mint(premium) → burn 50% + treasury 50%`

`sale → buyer(-100%) → seller(+net 97.5%) + treasury(+2.5%)`

`airdrop claim → user wallet · unclaimed → ecosystem`

`total minted on-chain now ≈ 605,138,750 BDNS · remaining ≈ 394,861,250 BDNS`

Token economy

Premium mints burn 50%

A short-name payment is split: half is burned (deflationary pressure on BDNS), half flows to the treasury that funds liquidity and chain operations.

Free tier, forever

5+ character domains cost nothing today and nothing tomorrow — a generous on-ramp that seeds the namespace with activity.

Marketplace fee

Resales pay a 2.5% fee, instantly split to the treasury. Secondary liquidity feeds the same engine as primary issuance.

Community launch allocation

The community allocation ("Community Staking", $10\% \cdot 100\text{M BDNS}$) is a Merkle-based claim — gasless bookkeeping, single-claim integrity, fully on-chain. The launch model:

Claimable at 12 months from TGE — no month-one flood, no instant-dump window.

Staking-first: the top-10 stakers by weight earn the protocol's staking rewards, so the allocation is earned with participation, not just claimed.

Unclaimed points roll back into the community staking pool — nothing is ever silently minted again; everything is reused on-chain.

The registry demo ships the same Merkle claim mechanism live on the L2 for hands-on testing; mainnet uses the identical contract layout.

🔗 The BDNS token

BDNS is the ERC-20 value layer of the chain: **supply cap 1B**, currently **~605.1M minted on-chain**. Its flows are exactly the protocol's flows.

🔗 Right now, on-chain

BDNS balance of buyer 0x8f86... — 1,000,000,000

BDNS allowance toward market — 0

Premium fee f treasury — BDNS balance 10,000.000 (0.4% of 250M)

🔗 Token economy

Utility: premium registrations and marketplace purchases are priced and settled in BDNS.

Deflation engine: premium mints burn 50% of the fee; the marketplace burns **1% for every 10 on-chain transactions**.

Permanent burn: the burn wallet has no known private key — its seed phrase was destroyed at genesis, making the burn irreversible.

Airdrop: claimable at 12 months with staking for the top-10 community stakers; unclaimed balances re-vest into the staking pool.

Treasury: fee revenue re-flows to liquidity, grants and chain incentives — the loop never stops.

🔗 The marketplace

The **BlockDNSMarketplace** escrows domains on-chain: a seller lists a token for a price in BDNS, the NFT moves into the contract, and any buyer pays directly — the fee splits to the seller and the treasury, and the domain transfers instantly. No intermediaries, no off-chain trust, no settlement risk.

🔗 Security & ownership

Domains are standard, battle-tested **ERC-721** tokens — owners keep custody in their own wallet.

The marketplace and airdrop use audited OpenZeppelin patterns: reentrancy guards, safe transfers and non-custodial escrow.

Everything — price, burn, treasury, sales — is public ledger data. Governance-free, deterministic and permissionless to verify.

This live build is an **open protocol demo** on a local L2; mainnet parameters mirror the same design.

Roadmap

Phase 1 — Naming core. Registry + pricer + BDNS + public dashboard. *Live in this demo.*

Phase 2 — Content. IPFS pinning and gateway resolution for every domain. *Live in this demo.*

Phase 3 — Liquidity. On-chain marketplace for domain resales. *Live in this demo.*

Phase 4 — Identity. Verified web3 handle — multi-chain records, ENS-style resolution DIDs.

Phase 5 — Scale. Mainnet deployment, bridge integration and multilingual UI.

This whitepaper describes the protocol as implemented in the running demo. Nothing in here is investment advice; token values accrue solely from protocol utility. Decentralize responsibly.